# Notes: Jacobson, LaLonde & Sullivan (AER, 1993)
## Earnings Losses of Displaced Workers

## Contents

# 1 Big picture

- **Question.** How large and persistent are the earnings losses experienced by high-tenure workers who lose jobs at distressed firms?
- **Core empirical setting.** The paper uses Pennsylvania administrative unemployment-insurance wage records linked to firm-level employment records, providing 52 quarters of earnings histories plus firm characteristics.
- **Main result.** High-tenure workers displaced from distressed firms experience long-term quarterly earnings losses that average roughly 25 percent of their predisplacement earnings.
- **Timing result.** Losses begin before final separation, drop sharply at separation, recover rapidly for only about six quarters, and then remain far below the counterfactual path.
- **Economic takeaway.** Displacement destroys firm-specific earnings components, match rents, wage premiums, deferred-compensation rents, and internal-labor-market advantages that are only slowly rebuilt.

# 2 Data and measurement

## 2.1 Pennsylvania administrative wage and firm records

The data combine a 5-percent sample of Pennsylvania workers from unemployment-insurance tax reports with ES202 firm-level employment records. Quarterly wage and salary earnings are observed from 1974 through 1986, and firm identifiers allow the authors to identify separations and firm employment declines.

**Detailed interpretation.** The key measurement advantage is that the authors observe both worker earnings paths and employer context.

## 2.2 High-tenure worker sample and displacement categories

The paper focuses on workers with six or more years of tenure by 1980, born between 1930 and 1959, with positive wage or salary earnings in each calendar year from 1974 to 1986. The sample has 9,507 separators, 13,704 stayers, 6,435 mass-layoff separators, and 3,072 non-mass-layoff separators.

**Detailed interpretation.** The mass-layoff sample is the most credible displacement sample because firm employment declines make voluntary quits or individual poor-performance explanations less plausible.

# 3 Models and empirical specifications

## 3.1 Definition of earnings loss

$$Loss_{it}(s) = E[y_{it} \mid D_{is} = 1, I_{i,s-p}] - E[y_{it} \mid D_{is} = 0, I_{i,s-p}].$$

**Detailed interpretation.** The loss compares actual earnings for workers who will be displaced to the counterfactual earnings path that would have occurred absent the events leading to displacement.

## 3.2 Baseline event-study model

$$y_{it} = \alpha_i + \gamma_t + \sum_{k \geq -m} D_{it}^k \delta_k + \varepsilon_{it}.$$

**Detailed interpretation.** Worker fixed effects remove permanent worker differences; time effects absorb aggregate shocks; event-time coefficients trace the displacement loss path.

### 3.3 Model with controls and worker-specific trends

$$y_{it} = \alpha_i + \gamma_t + x_{it}\beta + \tau_i t + \sum_{k \geq -m} D_{it}^k \delta_k + \varepsilon_{it}.$$

**Detailed interpretation.** Worker-specific trends test whether displaced workers were on slower earnings paths even before displacement.

### 3.4 Parsimonious heterogeneity model

$$y_{it} = \alpha_i + \gamma_t + x_{it}\beta + \sum_{k \geq -m} D_{it}^k \delta_k + F_{it}^1 c_i \phi_1 + F_{it}^2 c_i \phi_2 + F_{it}^3 c_i \phi_3 + \varepsilon_{it}.$$

**Detailed interpretation.** $F^1$ captures the pre-separation dip, $F^2$ the immediate post-separation drop, and $F^3$ the recovery slope after roughly six quarters.

## 4 Identification and empirical design

### 4.1 What the causal object is

The causal object is actual earnings relative to the counterfactual earnings path the worker would have followed absent the events leading to displacement. It includes forgone wage growth, pre-separation distress, and post-separation losses.

### 4.2 Main identifying comparison

Displaced workers are compared with nondisplaced high-tenure workers in the same administrative data system. Worker fixed effects absorb permanent earnings differences; calendar-quarter effects absorb statewide shocks; age-sex controls absorb life-cycle earnings patterns.

### 4.3 Why high tenure matters

High-tenure workers are most likely to have firm-specific capital, match quality, seniority, union rents, deferred compensation, or internal-labor-market positions. The sample therefore targets the group for whom displacement is expected to be most costly.

### 4.4 Why mass layoffs matter

Mass layoffs and plant closings are less likely to be caused by individual performance. Focusing on mass layoffs reduces selection concerns and better isolates firm-distress displacement.

### 4.5 Pre-trends as a diagnostic

The authors estimate effects as early as 20 quarters before separation. Losses are small four to five years before separation, begin about three years before separation, and accelerate near the layoff. This supports interpreting pre-separation losses as part of firm distress, not arbitrary worker selection.

### 4.6 Interpretation of pre-displacement losses

Pre-displacement losses may come from temporary layoffs, reduced hours, wage freezes, or weak wage growth. This means studies using one year before separation as the baseline understate total losses.

### 4.7 Why worker fixed effects matter

Table 1 shows stayers have higher 1979 earnings than separators despite similar ages. Fixed effects absorb this permanent level difference and prevent it from being mistaken for a displacement loss.

### 4.8 Why worker-specific trends matter

If displaced workers were simply on lower earnings-growth paths, losses would disappear after adding individual trends. They do not, so the main results are not driven by slow-growing workers being selected into displacement.

### 4.9 Same-firm comparison groups

Same-firm comparisons estimate the additional loss of separation relative to coworkers exposed to the same original-firm shock. For mass layoffs, estimates are smaller with same-firm stayers, implying that firm distress also hurts nondisplaced workers.

### 4.10 Local labor-market controls

Local labor-market trend, unemployment, and cyclical deviation help distinguish displacement losses from regional shocks. These controls affect heterogeneity but do not eliminate the large loss pattern.

### 4.11 Positive-earnings restriction

The requirement of positive annual earnings reduces attrition and missing-data problems, but may omit the worst nonemployment cases. Estimates are therefore best read as losses for workers who remain at least minimally attached to wage employment.

### 4.12 Remaining limitations

The design is not randomized, cannot fully separate wage cuts from hours cuts, and cannot completely distinguish firm-specific human capital from rents or internal-labor-market wage premiums.

## 5 Section-by-section study guide

- **Introduction:** motivates long-term earnings scarring and high-tenure workers.
- **Section I:** describes Pennsylvania UI and ES202 data and sample restrictions.

- **Section II:** defines earnings losses and lays out the event-study models.
- **Section III:** estimates the average path of losses and heterogeneity across worker, firm, industry, local-market, and reemployment-sector groups.
- **Conclusion:** connects the evidence to theories of firm-specific human capital, rent loss, internal labor markets, and implicit contracts.

# 6 Tables and figures

## 6.1 Table 1: Sample Characteristics + Figure 1: Quarterly Earnings of High-Attachment Workers Separating in Quarter 1982:1 and Workers Staying Through Quarter 1986:4

**Read:** Table 1 shows 9,507 separators and 13,704 stayers, similar age distributions, and somewhat lower 1979 earnings for separators. Figure 1 shows that workers separating in 1982:1 diverge from stayers before separation and remain far below them afterward.

**Detailed interpretation.** Table 1 establishes that the sample consists of high-tenure workers, not weakly attached workers. Figure 1 motivates the event-study approach by showing persistent raw earnings divergence.

**Economic message.** The first table and figure show that displacement is a persistent earnings-path shock, not just a temporary unemployment interruption.

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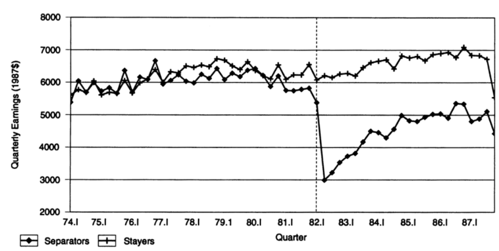
TABLE 1—SAMPLE CHARACTERISTICS

| Workers | Observations | Mean | Standard deviation | Median | 10th percentile | 90th percentile |
|---|---|---|---|---|---|---|
| A. *Age in 1979:* | | | | | | |
| Separators: | | | | | | |
| All | 9,507 | 37.0 | 7.4 | 37 | 27 | 47 |
| Males | 7,092 | 36.9 | 7.2 | 37 | 27 | 47 |
| Females | 2,415 | 37.3 | 7.8 | 38 | 27 | 48 |
| Nonmanufacturing | 2,870 | 36.9 | 7.3 | 37 | 27 | 47 |
| Manufacturing | 6,637 | 37.1 | 7.4 | 37 | 27 | 47 |
| Western Pennsylvania | 3,804 | 36.8 | 7.4 | 37 | 27 | 47 |
| Eastern Pennsylvania | 5,703 | 37.1 | 7.3 | 37 | 27 | 47 |
| Non-mass layoffs | 3,072 | 36.9 | 7.3 | 37 | 27 | 47 |
| Mass layoffs | 6,435 | 37.1 | 7.4 | 37 | 27 | 47 |
| Stayers | 13,704 | 37.7 | 7.0 | 38 | 28 | 47 |
| B. *1979 Earnings:* | | | | | | |
| Separators: | | | | | | |
| All | 9,507 | $24,196 | $12,287 | $22,904 | $11,525 | $36,798 |
| Males | 7,092 | 27,363 | 12,161 | 25,942 | 16,326 | 38,557 |
| Females | 2,415 | 14,897 | 6,641 | 14,275 | 7,595 | 22,928 |
| Nonmanufacturing | 2,870 | 24,648 | 15,547 | 22,363 | 10,029 | 39,358 |
| Manufacturing | 6,637 | 24,001 | 10,566 | 23,096 | 12,070 | 35,963 |
| Western Pennsylvania | 3,804 | 25,147 | 12,449 | 24,292 | 12,359 | 37,561 |
| Eastern Pennsylvania | 5,703 | 23,561 | 12,138 | 22,176 | 11,005 | 36,140 |
| Non-mass layoffs | 3,072 | 23,640 | 14,415 | 21,665 | 10,585 | 36,726 |
| Mass layoffs | 6,435 | 24,461 | 11,120 | 23,593 | 12,037 | 36,805 |
| Stayers | 13,704 | 26,322 | 12,980 | 24,867 | 13,644 | 38,880 |

(a) Table 1: Sample Characteristics




(b) Figure 1: Quarterly earnings

## 6.2 Figure 2: Earnings Losses for Separators in Mass-Layoff Sample

**Read:** Earnings losses begin about three years before separation, reach about 1,000$perquarterjustbeforeseparation, dropsh$
below expected levels five to six years later.

**Detailed interpretation.** Figure 2 is the paper's core result. It shows that displacement is a process and that the long-run cost is much larger than short-term unemployment.

**Economic message.** High-tenure displacement destroys durable earnings components.

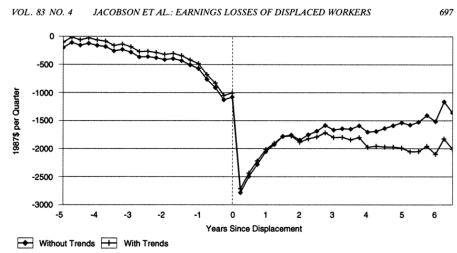


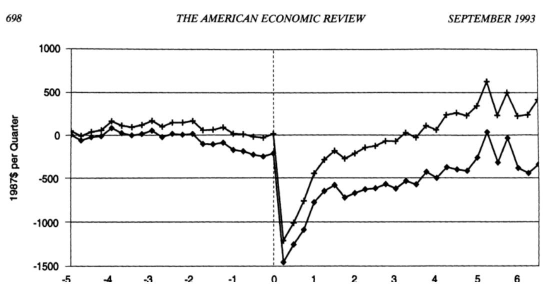

Figure 2: Earnings losses for separators in mass-layoff sample

## 6.3 Figure 3 and Figure 4

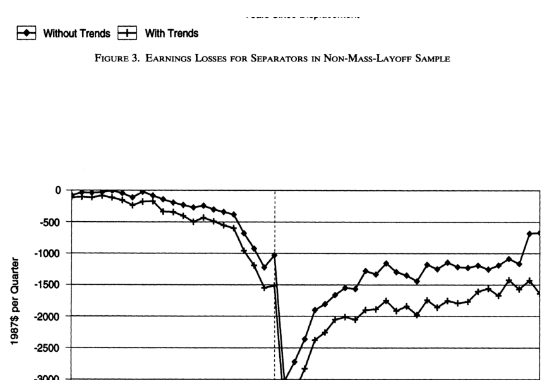
**Read:** Figure 3 shows smaller and less persistent losses for non-mass-layoff separators. Figure 4 shows that mass-layoff losses are about 20 percent smaller when stayers at the same firm are used as the comparison group, but losses remain large.

**Detailed interpretation.** The non-mass-layoff sample acts as a placebo-like comparison: the specification does not mechanically generate large persistent losses. The same-firm comparison shows that some distress affects stayers too, but displaced workers bear much larger costs.

**Economic message.** True displacement differs from ordinary separations.




(a) Figure 3: Non-mass-layoff losses




(b) Figure 4: Mass-layoff comparison groups

## 6.4 Figure 5: Sensitivity of Earnings-Loss Estimates for Non-Mass-Layoff Sample to Different Comparison Groups

**Read:** The two comparison-group estimates are very similar, showing that comparison-group choice matters less when the old firm is not in broad distress.

**Detailed interpretation.** This strengthens the interpretation that Figure 4 reflects real firm-wide distress effects, not model instability.

**Economic message.** Ordinary separations are not comparable to displacement from distressed firms.

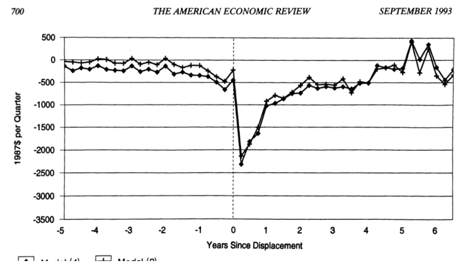



Figure 5: Non-mass-layoff comparison groups

## 6.5 Table 2: Losses by Worker Characteristics

**Read:** Table 2 decomposes losses into dip, drop, recovery, and fifth-year loss. Age and sex differences are modest, while industry, firm size, and local labor-market conditions matter more.

**Detailed interpretation.** The table supports mechanisms involving lost rents, union premia, local reemployment opportunities, and slow rebuilding of firm-specific capital.

**Economic message.** Displacement costs are broad-based, but their magnitude depends strongly on the old job and local labor market.

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TABLE 2—LOSSES BY WORKER CHARACTERISTICS

| | | Without other controls[a] | | | | | With other controls[b] | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Group | Number | Dip[c] | Drop[d] | Recovery[e] | Fifth-year loss dif | Fifth-year loss | Dip | Drop | Recovery | Fifth-year loss dif | Fifth-year loss |
| Overall | 6,435 | | | | | | −83.3 | −2,179 | 15.4 | — | −6,611 |
| | | | | | | | (2.2) | (16) | (4.4) | | (150) |
| Sex: | | | | | | | | | | | |
| Male | 4,972 | −10.8 | −217 | 6.5 | −545 | −7,143 | −3.4 | −103 | 4.7 | −177 | −6,788 |
| | | (0.7) | (7) | (0.9) | (40) | (132) | (0.7) | (7) | (0.9) | (43) | (157) |
| Female | 1,463 | 36.7 | 738 | −22.0 | 1,853 | −4,744 | 11.6 | 350 | −16.0 | 602 | −6,009 |
| | | (2.2) | (24) | (3.0) | (136) | (184) | (2.3) | (25) | (3.2) | (145) | (207) |
| Decade of birth: | | | | | | | | | | | |
| 1930's | 2,599 | −0.0 | 116 | −10.9 | −79 | −6,677 | −0.3 | 55 | −10.1 | −284 | −6,896 |
| | | (1.4) | (16) | (2.0) | (92) | (159) | (1.4) | (16) | (2.1) | (94) | (182) |
| 1940's | 2,584 | 7.2 | 3 | 4.6 | 241 | −6,356 | 3.6 | −28 | 5.6 | 171 | −6,440 |
| | | (1.4) | (15) | (2.0) | (87) | (151) | (1.4) | (15) | (2.0) | (88) | (172) |
| 1950's | 1,252 | −14.9 | −247 | 13.1 | −333 | −6,932 | −6.9 | −58 | 9.4 | 238 | −6,374 |
| | | (2.4) | (25) | (3.2) | (144) | (188) | (2.4) | (25) | (3.2) | (145) | (203) |
| Industry: | | | | | | | | | | | |
| Mining and construction | 247 | 1.3 | −497 | 7.5 | −1,616 | −8,435 | 9.5 | −387 | −0.1 | −1,549 | −8,160 |
| | | (5.6) | (58) | (7.6) | (332) | (352) | (5.8) | (59) | (7.8) | (339) | (369) |
| Nondurable manufacturing | 1,206 | 26.5 | 624 | −14.6 | 1,766 | −5,052 | 18.3 | 338 | −7.7 | 967 | −5,644 |
| | | (2.3) | (25) | (3.3) | (144) | (188) | (2.6) | (28) | (3.7) | (160) | (224) |
| Primary metals | 1,354 | −121.2 | −1,991 | 54.1 | −5,256 | −12,074 | −104.5 | −1,476 | 40.5 | −3,878 | −10,489 |
| | | (2.2) | (24) | (3.6) | (157) | (210) | (2.7) | (30) | (4.4) | (191) | (241) |
| Fabricated metals | 436 | 21.0 | 611 | −11.2 | 1,882 | −4,936 | 15.9 | 488 | −9.8 | 1,465 | −5,146 |
| | | (4.2) | (44) | (6.4) | (274) | (301) | (4.2) | (45) | (6.5) | (279) | (312) |
| Nonelectrical machinery | 632 | 47.9 | 1,005 | −36.9 | 2,174 | −4,644 | 35 | 797 | −27.4 | 1,817 | −4,794 |
| | | (3.4) | (38) | (5.8) | (249) | (284) | (3.5) | (39) | (5.9) | (257) | (306) |
| Electrical machinery | 421 | 43.2 | 288 | 7.0 | 1,500 | −5,318 | 49.5 | 494 | −2.7 | 1,842 | −4,769 |
| | | (4.2) | (46) | (6.1) | (270) | (300) | (4.3) | (47) | (6.4) | (282) | (322) |
| Transportation equipment | 419 | 25.0 | 422 | −27.5 | 310 | −6,508 | 14.1 | 215 | −15.5 | 85 | −6,526 |
| | | (4.3) | (46) | (6.2) | (264) | (291) | (4.4) | (48) | (6.6) | (282) | (324) |
| Other durable manufacturing | 441 | 25.6 | 525 | 3.0 | 2,248 | −4,570 | 18.9 | 338 | 9.1 | 1,807 | −4,804 |
| | | (4.2) | (43) | (5.5) | (237) | (262) | (4.2) | (43) | (5.7) | (242) | (282) |
| Transportation, communication, and public utilities | 348 | 6.6 | 150 | −63.5 | −2,573 | −9,392 | 5.5 | 66 | −63.6 | −2,916 | −9,527 |
| | | (4.7) | (49) | (7.0) | (295) | (321) | (4.8) | (50) | (7.1) | (301) | (333) |
| Wholesale and retail trade | 545 | 18.7 | 198 | 2.0 | 891 | −5,927 | 20.0 | 126 | 4.8 | 745 | −5,866 |
| | | (3.7) | (38) | (4.8) | (207) | (235) | (3.8) | (38) | (4.9) | (211) | (251) |
| Finance, insurance, and real estate | 183 | 127.7 | 1,312 | 14.3 | 5,963 | −855 | 115.7 | 947 | 24.3 | 5,004 | −1,608 |
| | | (6.6) | (70) | (8.2) | (352) | (369) | (6.7) | (72) | (8.3) | (358) | (387) |
| Professional, business, and entertainment services | 203 | 82.0 | 1,158 | −18.2 | 3,725 | −3,093 | 93.1 | 1,270 | −26.2 | 3,769 | −2,843 |
| | | (6.3) | (63) | (8.4) | (360) | (378) | (6.4) | (64) | (8.7) | (369) | (394) |
| Firm size: | | | | | | | | | | | |
| 50–500 | 1,704 | 7.9 | 351 | 0.6 | 1,434 | −5,403 | −16.1 | −37 | 13.0 | 501 | −6,110 |
| | | (1.9) | (20) | (2.6) | (113) | (163) | (2.1) | (22) | (2.9) | (124) | (193) |
| 501–2,000 | 1,497 | 33.5 | 501 | −14.1 | 1,298 | −5,540 | 13.9 | 214 | −4.7 | 625 | −5,986 |
| | | (2.0) | (22) | (2.9) | (127) | (176) | (2.2) | (23) | (3.1) | (135) | (246) |
| 2,001–5,000 | 1,381 | 40.9 | 720 | −32.3 | 1,267 | −5,570 | 27.2 | 480 | −23.8 | 730 | −5,881 |
| | | (2.2) | (23) | (3.1) | (134) | (179) | (2.3) | (24) | (3.5) | (149) | (203) |
| Greater than 5,000 | 1,853 | −64.8 | −1,265 | 34.9 | −3,312 | −10.150 | −16.7 | −497 | 9.6 | −1,510 | −8,121 |
| | | (1.8) | (19) | (2.9) | (125) | (190) | (2.3) | (25) | (3.6) | (154) | (224) |



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TABLE 2—*Continued.*

| | | Without other controls[a] | | | | | With other controls[b] | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Group | Number | Dip[c] | Drop[d] | Recovery[e] | Fifth-year loss dif | Fifth-year loss | Dip | Drop | Recovery | Fifth-year loss dif | Fifth-year loss |
| Local labor market: | | | | | | | | | | | |
| Employment trend | | 129.2 | 2,391 | −73.2 | 5,903 | | 38.8 | 743 | 18.1 | 2,069 | |
| | | (7.5) | (84) | (10.9) | (485) | | (7.9) | (87) | (11.7) | (520) | |
| Employment deviation | | 79.9 | 900 | −50.3 | 1,082 | | 51.8 | 376 | −40.9 | −540 | |
| | | (6.4) | (64) | (9.3) | (401) | | (6.4) | (65) | (9.4) | (408) | |
| Unemployment rate | | 41.5 | 213 | −29.7 | −635 | | 1.2 | −555 | 1.3 | −2,153 | |
| | | (8.9) | (97) | (14.3) | (631) | | (9.0) | (98) | (14.6) | (643) | |

*Notes:* "Dip," "drop," "recovery," and "fifth year loss diff" columns give groups' deviations from the mean of the variable for all displaced workers given in the first row of the table. See text for full explanations of entries. Numbers in parentheses are standard errors.
[a]Estimates derived from models that include interactions with exactly one of sex, birth cohort, industry, firm size, or local labor-market interactions.
[b]Model includes all interactions with sex, birth cohort, industry, firm size, and local labor market.
[c]Coefficient on predisplacement time trend.
[d]Coefficient on dummy for first six quarters after displacement.
[e]Coefficient on postseparation time trend.

that the differences between these groups' patterns of losses are generally very small. Younger workers have a somewhat greater rate of decline in the period before separation, probably reflecting a greater vulnerability to temporary layoffs due to lower lev-

are substantial across a broad range of industries and firm sizes. The pattern of losses for workers displaced from industries as diverse as nondurable manufacturing, motor vehicles, and wholesale and retail trade closely resembles the pattern depicted in

### 6.6 Table 3: Earnings Losses by Sector of New Job

**Read:** Manufacturing workers who leave manufacturing lose about 38 percent of predisplacement earnings at 24 quarters; even workers who return to the same four-digit SIC lose about 20 percent. Nonmanufacturing workers show similar persistent losses.

**Detailed interpretation.** Returning to the same industry does not eliminate the loss, implying that firm-specific components of earnings are important.

**Economic message.** Reemployment is not enough; even similar-job reemployment leaves large earnings scars.

TABLE 3—EARNINGS LOSSES BY SECTOR OF NEW JOB: DEVIATION BETWEEN ACTUAL AND EXPECTED QUARTERLY EARNINGS

| | New job in same sector | | |
|---|---|---|---|
| Quarters since separation | Same four-digit SIC | Different four-digit SIC | New job in other sector |
| A. *Displaced Manufacturing Workers:* | | | |
| −8 | −$379 | −$117 | −$237 |
| | (82) | (67) | (73) |
| | [−7] | [−2] | [−4] |
| 12 | −1,044 | −1,117 | −2,616 |
| | (82) | (67) | (73) |
| | [−19] | [−21] | [−44] |
| 24 | −1,103 | −958 | −2,221 |
| | (197) | (137) | (150) |
| | [−20] | [−18] | [−38] |
| B. *Displaced Nonmanufacturing Workers:* | | | |
| −8 | −229 | −26 | −151 |
| | (132) | (128) | (231) |
| | [−4] | [0] | [−3] |
| 12 | −1,129 | −1,305 | −1,498 |
| | (132) | (128) | (231) |
| | [−18] | [−23] | [−26] |
| 24 | −1,103 | −1,276 | −1,949 |
| | (315) | (241) | (476) |
| | [−18] | [−22] | [−33] |

*Notes:* Numbers in parentheses are standard errors. Numbers in square brackets express the estimated losses as a percentage of predisplacement earnings.

percent if they found new manufacturing

Table 3: Earnings losses by sector of new job

## 7 Short summary

- Mass-layoff separators suffer long-term losses of roughly 25 percent of predisplacement earnings.
- Losses begin before separation, drop sharply at separation, and recover only partially.
- Non-mass-layoff separators have smaller and less persistent losses.
- Age and sex differences are modest; industry, firm size, and local labor markets matter more.
- Even workers reemployed in the same industry suffer large losses.

# 8 Conclusion

## 8.1 Core conclusion

Displacement for high-tenure workers creates large, persistent earnings losses that extend far beyond the initial spell of unemployment.

## 8.2 Economic intuition

Long-term jobs contain firm-specific capital, match rents, wage premiums, and internal-labor-market advantages that are hard to replace once the employment relationship is destroyed.

## 8.3 Why pre-separation declines matter

Displacement is a process, not just a date. Earnings begin falling before final separation, so short pre-periods understate true losses.

## 8.4 Policy implications

The evidence supports policy responses beyond unemployment insurance, including wage insurance, retraining, mobility assistance, and local labor-market support.

## 8.5 Limitations and open questions

The paper cannot fully distinguish lower wages from lower hours, cannot observe all out-of-state or self-employment outcomes, and cannot fully decompose losses into skill loss, rent loss, or internal-labor-market mechanisms.

## 8.6 Takeaway

The economic cost of structural change is much larger than immediate joblessness; for many high-tenure workers, displacement is a long-run earnings shock.